

Mid-zone hepatocytes trade proliferation for survival via Atf4-Chop axis in early acute liver injury

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eLife Assessment

This study addresses an **important** question in liver biology: how zonal hepatocytes balance survival and proliferation following injury; using spatial transcriptomics, mechanistic perturbations, and functional assays, the authors propose that a mid-zone Atf4-Chop axis to Btg2 program temporarily suppresses proliferation to promote survival during APAP-induced hepatotoxicity. The idea that distinct intrahepatic zones mount tailored stress responses is conceptually significant and has implications for regeneration and toxicology. The dataset is rich and the methodology modern, but several conclusions rely on assumptions about zonation under injury, limited injury models, and **incomplete** functional validation of the Atf4-Chop-Btg2 axis. With targeted revisions and additional experiments, the work has the potential to provide strong mechanistic insights into liver zonation and injury responses.

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Abstract

Hepatocytes undergo extensive proliferation to facilitate liver repair after injury, yet early adaptive changes prior to proliferation remain unclear. Here, we report that during early acetaminophen (APAP)-induced liver injury (AILI), hepatocytes exhibit transient proliferation suppression, most pronounced in mid-zone hepatocytes due to zonal APAP metabolism. Using spatial transcriptomics, immunohistochemistry, and functional studies, we identified a unique mid-zone stress-response program. Central to this adaptation is the Atf4-Chop axis, which actively suppresses proliferation via the cell cycle inhibitor Btg2, prioritizing cytoprotection over cell division. This transient arrest is a critical survival strategy: halting energy-intensive proliferation during peak injury allows mid-zone hepatocytes to redirect resources towards protection, enhancing their survival in early AILI. Thus, Atf4-Chop-mediated quiescence preserves a hepatocyte reservoir necessary for subsequent regenerative proliferation and effective repair. Our findings reveal a key adaptive trade-off in mid-zone hepatocytes where transient proliferation arrest promotes early survival to enable repair.

Introduction

Acetaminophen (APAP) overdose is the leading cause of drug-induced acute liver failure worldwide, resulting in over 10,000 hospitalizations and approximately 500 deaths annually in the United States.^{1–4} The hepatotoxicity of APAP is initiated by its metabolic activation through cytochrome P450 enzymes, primarily in pericentral hepatocytes, generating the reactive metabolite N-acetyl-p-benzoquinone imine (NAPQI).^{5–8} NAPQI depletes glutathione and forms protein adducts, leading to oxidative stress and mitochondrial dysfunction that culminate in centrilobular necrosis.^{5–8} Although the mechanisms of APAP bioactivation and toxicity are well-characterized, the early adaptive responses of hepatocytes—particularly their proliferative dynamics across lobular zones—remain poorly understood.

The liver is organized into functional hexagonal units termed lobules, where hepatocytes display spatial heterogeneity across three distinct zones: periportal (PP, zone 1), mid-zonal (Mid, zone 2), and pericentral (PC, zone 3).^{9–11} These zones exhibit unique gene expression profiles and metabolic specializations.⁹ Pericentral hepatocytes (zone 3), expressing high levels of cytochrome P450 enzymes, are the primary site of acetaminophen (APAP)-induced necrosis.¹⁰ However, recent spatial profiling, functional, and lineage tracing studies have identified midzonal hepatocytes (zone 2) as the predominant source of new hepatocytes during both homeostasis and repair, demonstrating significant proliferative capacity and metabolic plasticity.^{12–14} Despite these advances, how mid-zone hepatocytes dynamically adapt during the initial phase of acute stress remains unknown.

A key regulator of cellular stress adaptation is the integrated stress response (ISR), which phosphorylates eukaryotic initiation factor 2 α (eIF2 α) to suppress global translation while selectively upregulating stress-responsive genes, including activating transcription factor 4 (Atf4) and its downstream target pro-apoptotic factor C/EBP homologous protein (Chop, encoded by DNA damage-inducible transcript 3 [Ddit3]).^{15,16} Although Chop is traditionally associated with apoptosis, its transcriptional targets exhibit significant overlap with those of Atf4, encompassing genes that paradoxically support cell survival and growth.^{17–19} These findings underscore the context-dependent duality of the Atf4-Chop axis across different stress conditions, highlighting the need to elucidate its precise regulatory roles in specific physiological settings.

Through spatial transcriptomics (ST) and functional analyses, we demonstrate that zonal heterogeneity in APAP metabolism drives a stress-adaptive response in mid-zone hepatocytes during early AILI. This response is mediated by the Atf4-Chop-Btg2 axis, which orchestrates a transient proliferation arrest to prioritize cellular survival over regenerative capacity. Our findings elucidate the spatial and molecular mechanisms governing hepatocyte adaptation to acute injury, uncovering a critical survival-regeneration trade-off in early phase of acute liver injury.

Results

Mid-zone hepatocytes exhibit the most pronounced proliferative decline during early acute liver injury

To assess hepatocyte proliferation in early AILI, wild-type C57BL/6J mice were intraperitoneally injected with 300 mg/kg APAP, and liver sections were analyzed at multiple time points (0, 3, 6, 12, and 24 hours post-APAP). Immunohistochemical staining for Ki67, a proliferation marker, revealed that mid-zone hepatocytes exhibited a roughly 3-fold higher basal proliferative activity than PP

and PC hepatocytes under homeostatic conditions (0 hour) (**Figure 1A**, B). During the early injury phase (3–12 hours post-APAP), the number of Ki67⁺ hepatocytes in the mid-zone sharply declined from 70 to 20 per section, reaching proliferation levels similar to those in PP and PC zones (**Figure 1A**, B). By 24 hours post-APAP, mid-zone proliferation gradually recovered (**Figure 1A**, B), indicating a transient suppression of proliferation in this region during early AILI. Further analysis of proliferation dynamics in each zone confirmed that all zones exhibited reduced proliferation during the injury initiation phase, with the mid-zone showing the most decline (**Figure 1C**). Comparative value change analysis (relative to 0 hour) further supported these findings, highlighting the mid-zone as the most affected region in terms of proliferative suppression during early AILI (**Figure 1D**).

Mid-zone hepatocytes show distinct gene expression profiles with reduced proliferative activity during early acute injury

To investigate the mechanism underlying proliferation pause during early AILI, liver sections from wild-type mice treated with APAP for various time points were subjected to ST using the Visium (10 × Genomics) platform (**Figure S1A**). Due to variability in mice's responses to APAP, multiple mice were treated at each time point, and the samples representing the average level at each time point were selected based on Hematoxylin and eosin (H&E) staining and serum alanine aminotransferase (ALT) levels for ST analysis. H&E staining of liver sections used for ST revealed a gradual increase in the necrotic area from 0 to 24 hours (**Figure S1B**). Spatial map analysis consistently and unbiasedly divided the liver into three zones (PP, Mid, and PC) across all time points (**Figures S1C**). Analysis of ST data at 0, 3, 6, and 24 hours post-APAP via UMAP plots consistently revealed distinct zonal clusters and dynamic spatiotemporal patterns during the early phase of acute injury (**Figures S1D** and **S1E**). Spatiotemporally resolved heatmaps of representative markers, such as *Glul* (a PC marker) and *Cyp2f2* (a PP marker), further reinforced the stability of zonation profiles during early acute liver injury (**Figure S1F**). These findings suggest that while gene expression profiles change during acute injury, the fundamental spatial architecture of liver zones remains stable. Additionally, immunofluorescence staining of Glutamine Synthetase (GS, the protein product of *Glul*) and RNA-FISH for *Cyp2f2* in liver sections collected at 6 hours post-APAP confirmed these ST analyses (**Figure S1G**).

We further analyzed differentially expressed genes (DEGs) in hepatocyte zones (PP, Mid, and PC) at 0, 3, 6, and 24 hours post-APAP (**Figure 2A**). Throughout the 0 to 24-hour period post-APAP, gene expression in the PP remained relatively stable. At 0, 3, and 6 hours post-APAP, genes associated with metabolism (such as *Hsd17b13*, *Fabp1*, *Apoc2*, *Thrsp* for fat metabolism; *Rida*, *Hal*, *Gls2*, *Bhmt* for fatty acid metabolism; *Aldob*, *Pck1* for glucose metabolism) and transport (such as *Alb*, *Hpx*, *Selenop*, *Trf*, *Lcn2*) were prominently expressed, with a notable upregulation of inflammatory response-related genes (such as *Orm1*, *Serpina1c*, *Lrg1*) observed at 24 hours post-APAP (**Figure 2A**). In contrast, Mid-zone exhibited significant changes over time. Initially, it showed high expression of genes related to cell proliferation (such as *Igf1bp2*), metabolism (such as *Scd1*, *Cdo1*, *Ces1c*, *Ttr*, *Apoa1*), and homeostasis (such as *Hamp*, *Cps1*, *Rgn*) at 0 hours, transitioning to stress response/cell survival (such as *Hmox1*, *Hspa8*, *Atf3*, *Osgin1*) and protein degradation/autophagy (such as *Ubb*, *Ubc*, *Sqstm1*) at 3 and 6 hours, and eventually shifting to cell proliferation and wound healing (such as *Fgg*, *Fgl1*, *Fgb*) by 24 hours post-APAP (**Figure 2A**). The PC showed a decrease in xenobiotic metabolism (such as *Cyp3a11*, *Cyp2e1*, *Cyp1a2*, *Cyp2c37*) and fat metabolism (such as *Apoe*, *Akr1c6*, *Hpd*, *Oat*) during early AILI (at 0, 3, 6 hours post-APAP), with a marked increase in the expression of genes related to cytoskeleton (such as *Actb*, *Krt8*, *Krt18*) and redox regulation (such as *Prdx1*, *Srxn1*) observed at 24 hours post-APAP (**Figure 2A**). These differential expression patterns across various zones and time points post-APAP highlight significant changes in Mid-zone. Notably, our analysis suggests that the Mid-zone undergoes a stress adaption during the initial phases of acute liver injury.

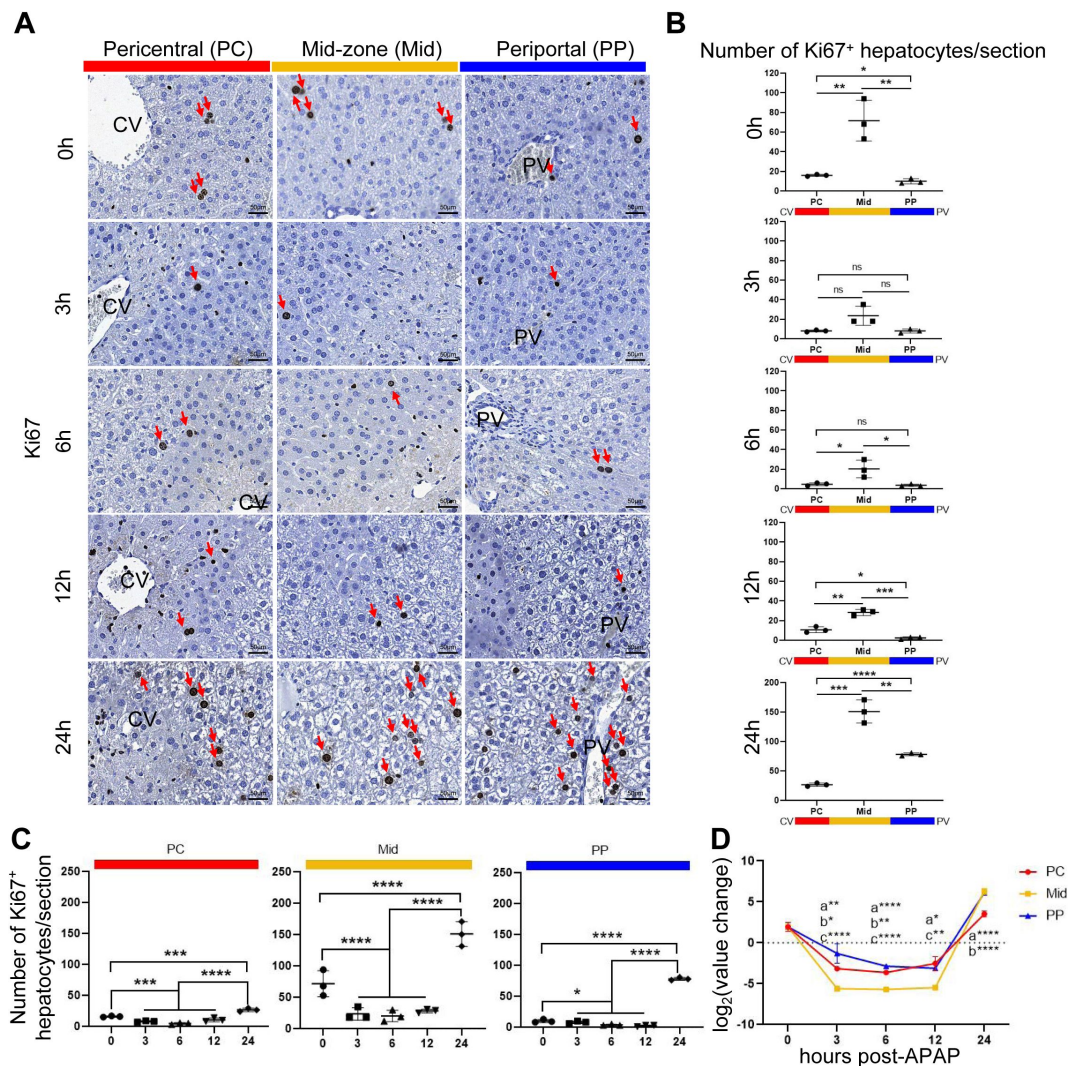


Figure 1.

Mid-zone hepatocytes exhibit the most pronounced proliferative decline in early acute liver injury.

Wildtype C57BL/6J mice were intraperitoneally injected with 300 mg/kg APAP for 0, 3, 6, 12 and 24 h. **(A)** Immunohistochemical staining of Ki67 to detect proliferating hepatocytes in liver sections at 0, 3, 6, 12 and 24 h post-APAP. Representative images of pericentral (PC), middle (Mid) and periportal (PP) are shown. Red arrows indicate Ki67-positive hepatocytes. **(B)** Quantitative analysis of Ki67⁺ hepatocytes across liver zones (PC, Mid, PP) at 0, 3, 6, 12, and 24 h post-APAP. **(C)** Quantitative analysis of Ki67⁺ hepatocytes across 0, 3, 6, 12, and 24 h post-APAP in each liver zone (PC, Mid, PP). **(D)** Changes of Ki67-positive cells across liver zones (PC, Mid, PP) at 0, 3, 6, 12 and 24 h post-APAP. Data are presented as log₂-value change (number of Ki67-positive cells at x h post-APAP – the number of Ki67-positive cells at 0 h post-APAP). (a) denotes significance between PC and Mid regions, (b) denotes significance between PC and PP regions, and (c) denotes significance between Mid and PP regions. Data points represent mean hepatocyte counts from six high-power fields per zonal layer (n=3 mice/group) in B and C. Data are represented as means ± SD; *p < 0.05; **p < 0.01; ***p < 0.001; ****p < 0.0001; ns, not significant.

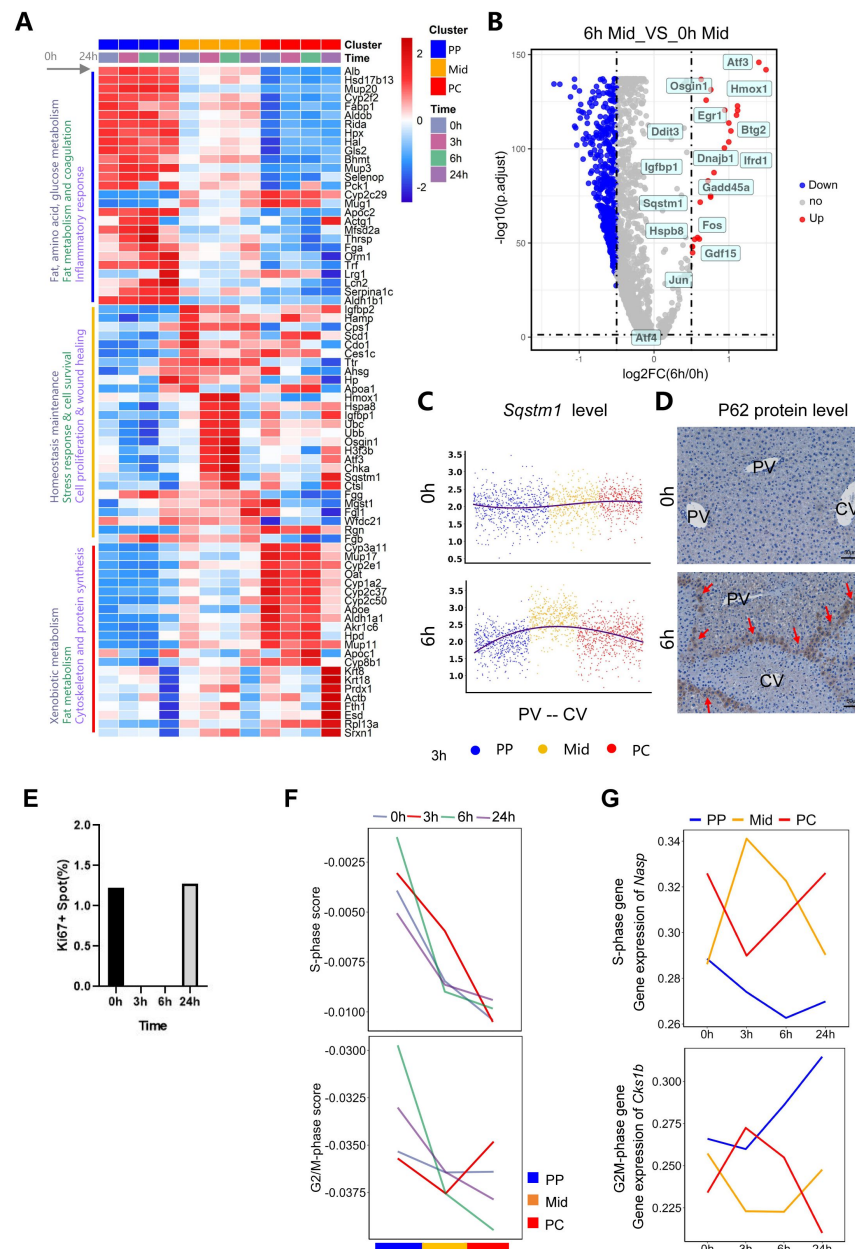


Figure 2.

Mid-zone hepatocytes show distinct gene expression profiles with reduced proliferative activity during early acute injury.

(A) Heatmap showing the gene expression of the top ten differentially expressed genes in hepatocyte zones (zones PP, Mid, and PC) at 0, 3, 6, 24 h post-APAP. The major functions of hepatocytes in each zone are labeled on the left side for 0, 6, and 24 h post-APAP, respectively. (B) Volcano plot illustrating the DEGs in Mid-zone at 6 h compared to 0 h post-APAP. Gray dots denote genes that are not statistically significant. Red and blue dots represent genes that are upregulated and downregulated, respectively, in the sample tissue, at least a 0.5-fold difference from the matched control, with a false discovery rate (FDR) threshold of 0.05. (C) Scatter plot shows the dynamic changes of *Sqstm1* mean expression level in hepatocyte zones (zones PP, Mid, and PC) at 0 and 6 h post-APAP. (D) Immunohistochemical staining of p62 (protein product of *Sqstm1*) to detect the dynamic changes of p62 across hepatocyte zones at 0 and 6 h post-APAP. Red arrows indicate p62-positive hepatocytes. (E) Percentage of Ki67-positive spots analyzed at 0, 3, 6 and 24 h post-APAP. (F) Average module score for S-phase (up) and G2/M-phase (down) genes across hepatocyte zones (zones PP, Mid, and PC) at 0, 3, 6 and 24 h post-APAP. (G) Expression levels of the S-phase gene *Nasp* (up) and the G2/M phase gene *Cks1b* (down) in each hepatocyte zone (zones PP, Mid, and PC) at 0, 3, 6, 24 h post-APAP.

Further analysis focuses on DEGs in Mid-zone at 6 hours post-APAP compared to 0 hour. The volcano plot revealed a notable increase in the number of genes significantly upregulated in the mid zones at 6 hours, predominantly associated with the stress response and adaptation (such as *Atf3*, *Hmox1*, *Osgin1*, *Ddit3*, *Gadd45a*, *Atf4*, *Jun*, *Hspa1*) as well as cell proliferation and apoptosis (such as *Btg2*, *Egr1*, *Igf1bp1*, *Sqstm1*, *Dnajb1*) (**Figure 2B**). Among the significantly upregulated genes, *Sqstm1* (Sequestosome 1) was selected for further validation. *Sqstm1* encodes the p62 protein, a key regulator of autophagy and oxidative stress response.²⁰ While *Sqstm1* expression was almost evenly distributed across all zones at 0 hours, it was markedly elevated in the mid-zone at 6 hours post-APAP (**Figure 2C**). Immunohistochemical staining further confirmed the distinct localization of p62 in the mid-zone at this time point (**Figure 2D**). Together, our findings indicate that Mid-zone hepatocytes undergo dynamic changes during the early phases of acute liver injury, primarily characterized by stress response and adaptation.

Further ST analysis revealed that Ki67-positive spots also show a transient reduction during the early injury (at 3 and 6 hours post-APAP) (**Figure 2E**), which aligns with the findings from Ki67 immunohistochemical staining. Cell cycle analysis is crucial for understanding cell proliferation, as it reviews the stages of the cell cycle that cells are in and how many cells are actively dividing. (23) In the cell cycle, the S phase is responsible for DNA synthesis and replication, while the G2M phase prepares cells for and executes mitosis.(23) Analysis of cell-cycle phase distribution, based on genes identified in previous studies,(24, 25) revealed that Mid-zone hepatocytes exhibited a transient increase in S-phase genes at 3 hours, followed by a decline at later time points (6, and 24 hours). In contrast, Mid-zone hepatocytes exhibited reduced G2/M-phase scores at 3 and 6 hours, returning to baseline levels by 24 hours (**Figure 2F**). This suggests that Mid-zone hepatocytes may initiate DNA synthesis, but they are unable to complete the cycle due to the environmental toxicity or cellular damage. Nuclear Autoantigenic Sperm Protein (*Nasp*) binds to histones and facilitate chromatin assembly during the S phase.²¹ Casein Kinase 1 Beta (*Cks1b*) participates in the control of G1 phase and the transition to the S phase.²² High *Nasp* levels indicate efforts toward DNA repair and chromatin stability, while low *Cks1b* levels point to a potential cell cycle arrest or a reduction in cell proliferation to manage the damage (**Figure 2G**). Overall, ST analysis revealed that Mid-zone hepatocytes exhibit distinct gene expression profiles and cell cycle arrest during early acute injury.

Zonal metabolism of APAP leads to stress response in Mid-zone hepatocytes

Next, we aimed to elucidate the cause for the stress response observed in the Mid-zone during early APAP. APAP is metabolized by cytochrome P450 enzymes (Cyp family) into reactive metabolites, particularly N-acetyl-p-benzoquinone imine (NAPQI), which can induce oxidative stress and damage cellular components if not promptly detoxified by glutathione (GSH) (**Figure 3A**). We analyzed the gene expression levels of Cyp family enzymes in hepatocyte zones (PP, Mid, and PC) at 0, 3, 6, and 24 hours post-APAP. Our analysis revealed a gradient decrease in the expression levels of most Cyp enzymes, showing a trend from high expression in PC to lower expression in PP (**Figure 3B**). Scatter plots of representative Cyp enzymes, such as *Cyp2e1* and *Cyp1a2*, clearly showed a temporal and spatial expression pattern consistent with this gradient decrease from CV to PV (Figure S2A). Immunohistochemical staining of *Cyp1a2* confirmed its gradient decrease in protein expression from the CV to the PV under homeostatic conditions (0 hour). However, following APAP-induced toxicity, *Cyp1a2* expression declined around the CV—where necrosis was prominent—during the injury initiation phase (3–24 hours post-APAP). Consequently, the Mid-zone adjacent to necrotic regions emerged as the primary area retaining *Cyp1a2* expression (**Figure 3C**, D). These findings suggest that the spatial redistribution of Cyp expression establishes the Mid-zone as the primary site of APAP metabolism during early AILI, thereby inducing subsequent transcriptional reprogramming (**Figure 3E**).

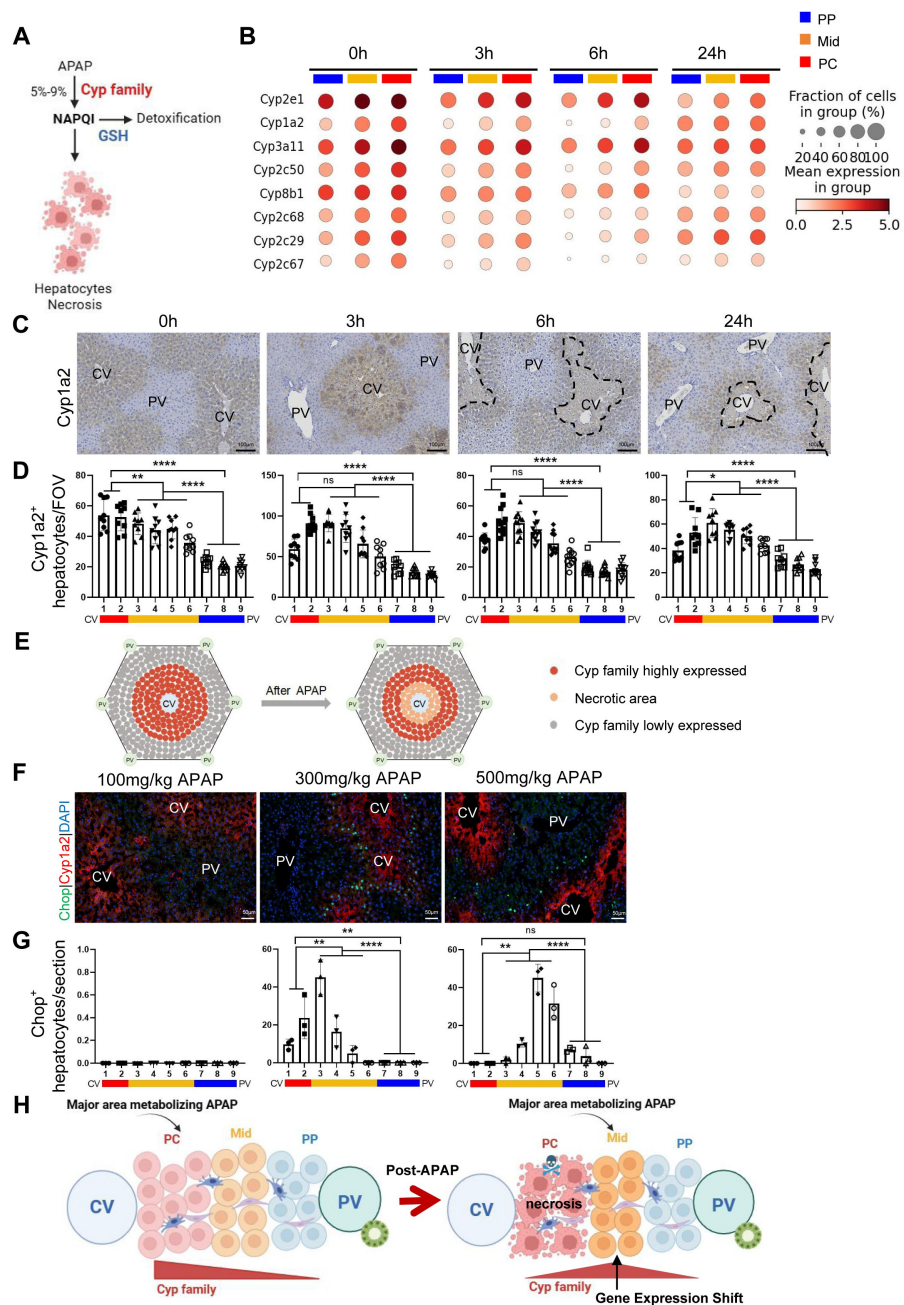


Figure 3.

Zonal metabolism of APAP leads to stress response in Mid-zone hepatocytes.

(A) Schematic figure illustrating the metabolic process of liver injury caused by APAP. (B) Dot plot displays the scaled gene expression levels of the Cyp family in hepatocyte zones (zones PP, Mid, and PC) at 0, 3, 6, 24 h post-APAP. (C) Immunohistochemical staining of Cyp1a2 to detect its dynamic changes in hepatocyte zones (from CV to PV) at 0, 3, 6, 24 h post-APAP. Injured area is outline by black dashed lines. (D) Quantification of zonal distribution of Cyp1a2-positive cells in liver sections at 0, 3, 6, and 24 h post-APAP, respectively. (E) Schematic figure illustrating expression changes of Cyp1a2 before and after APAP. (F) Immunofluorescence staining of Chop protein (green) to observe the dynamic changes in its expression across hepatocyte zones (from CV to PV) at 6 h post-APAP with doses of 100, 300, or 500 mg/kg. Cyp1a2 protein (red) staining highlights the area around the CV. Cell nuclei are stained with DAPI (blue). (G) Quantification of zonal distribution of Chop-positive cells in liver sections from mice treated with varying doses of APAP– 100, 300, or 500 mg/kg. (H) Schematic figure illustrating zonal metabolism of APAP results in the gene expression shift in Mid hepatocytes following necrosis. Data are represented as means $\pm$ SD; * p < 0.05; ** p < 0.01; *** p < 0.001; **** p < 0.0001; ns, not significant.

To investigate this hypothesis, we administered mice with varying doses of APAP (100, 300, and 500 mg/kg) to induce different extents of liver injury. H&E staining revealed a dose-dependent increase in the size of the injured areas, with higher doses resulting in larger areas of injury (Figure S2B and S2C). Subsequently, we performed immunofluorescence staining for Chop, a protein differentially highly expressed in the Mid-zone at 6 hours post-APAP compared to 0 hour (as identified in **Figure 2B**), to assess the Mid-zone response. Our findings indicate that Chop expression is absent at low doses of APAP when necrosis does not occur. However, as the dose of APAP increases, the injured area expands, and Chop expression shifts toward the PV, consistently surrounding the injured area (**Figure 3F and G**). Together, these findings demonstrate that CV necrosis establishes the adjacent peri-necrotic region as the dominant site of APAP metabolism, contingent upon both Cyp family expression levels and APAP dosage. This spatially restricted metabolic adaptation drives subsequent region-specific transcriptional changes in the surrounding parenchyma (**Figure 3H**).

The Atf4-Chop axis emerges as pivotal in Mid-zone during early acute injury

To explore master regulator of stress adaptation in Mid-zone, we constructed gene regulatory networks from the ST data. We identified the top 10 regulons in each zone at 3 and 6 hours post-APAP (**Figure 4A** and Figure S3A). A closer examination of the transcription factors (TFs) in the Mid-zone revealed that *Atf4*, *Fos*, *Nfil3*, *Egr1*, *Maff*, *Atf3*, and *Ddit3* co-activated at both time points. *Atf4*, *Chop*, and *Atf3* play crucial roles in managing cellular stress responses.^{23,24} *Fos* and *Egr1* are immediate early genes that are rapidly activated in response to stimuli, regulating genes related to growth and stress responses.²⁵ *Maff* and *Nfil3* contribute to the management of oxidative stress and metabolic processes.^{26,27} The elevated activity of these TFs in the Mid-zone indicates that this region is undergoing a complex response to stress, serving as a critical area for cellular adaptation, survival, and potential necrosis during early liver injury.

Among the top 10 TFs with highest activity in the Mid-zone during early AILI, *Ddit3* and *Atf3* were the two most highly expressed, with *Atf4* ranking seventh (**Figure 4B** and Figure S3B). Since *Ddit3* expression is often regulated by *Atf4*, and the *Atf4*-*Ddit3* axis plays a critical role in stress adaptation,²⁸ we chose to focus on this pathway for further investigation. The violin plot analysis demonstrated that *Atf4* and *Ddit3* were prominently active in Mid-zone, particularly at 3 and 6 hours post-APAP (**Figure 4C** and Figure S3C). Gene regulatory networks illustrated associations between *Atf4* and its target genes (such as *Gdf15*, *Hspa8*, and *Ddit3*) and *Ddit3* with its targets (such as *Hspa8*, *Hmox1*, *Osgin1*, *Gadd45a*, and *Egr1*) at 3 hours post-APAP (Figure S3D). Heatmap further clearly showed that the target genes of *Atf4* and *Ddit3* are exclusively expressed in the Mid-zone hepatocytes at 6 hours post-APAP (**Figure 4D**). Many of these target genes were DEGs in Mid-zone during early AILI, underscoring the critical role of the *Atf4*-*Ddit3* axis in orchestrating gene expression changes in this region.

From the regulatory networks, it is evident that many genes regulated by *Atf4* and *Chop* overlap. Therefore, we conducted a GO pathway analysis to identify the pathways co-regulated by *Atf4* and *Chop*. The results revealed the top 15 enriched pathways for genes co-regulated by *Atf4* and *Chop* in Mid-zone at 3 and 6 hours post-APAP (**Figure 4E** and Figure S3E). These pathways encompass responses to unfolded proteins, stress, autophagy, and regulation of apoptotic signaling pathways. Additionally, GO pathway analysis identified the top 15 enriched pathways for up-regulated genes specifically in Mid-zone at 3 hours post-APAP compared to other zones, further highlighting pathways co-regulated by *Atf4* and *Chop* in response to APAP (Figure S3F). The pathways co-regulated by *Atf4* and *Chop* point to the activation of ISR. To confirm the occurrence of ISR in Mid-zone during the initiation stage of AILI, we performed immunohistochemical staining for phosphorylated eIF2 α (p-eIF2 α) at 0 and 3 hours post-APAP and observed increased expression of p-eIF2 α in the mid-zone (**Figure 4F**). While the ISR is a generalized pathway integrating endoplasmic reticulum (ER) stress and other insults through eIF2 α kinases, ER stress is a subset of

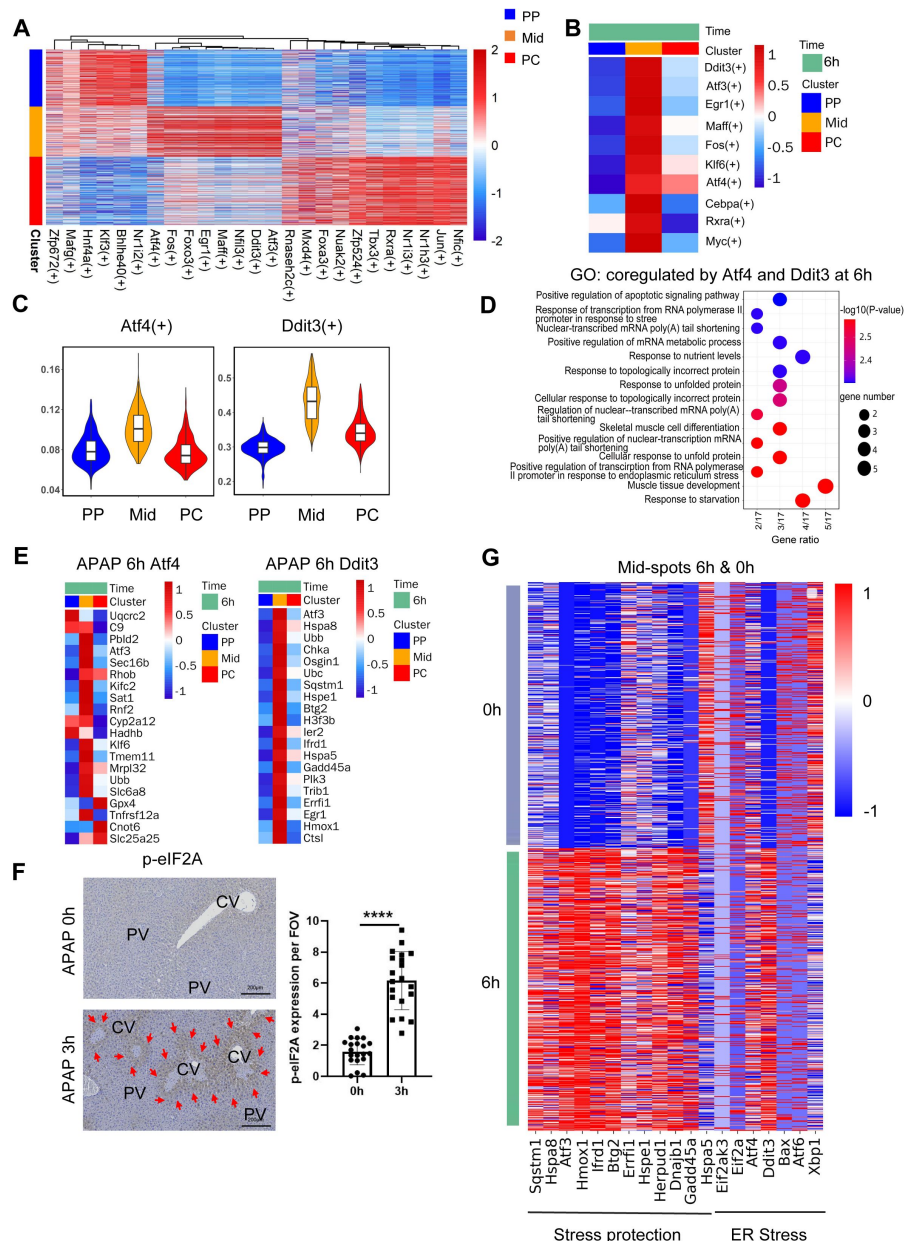


Figure 4.

The Atf4-Ddit3 axis emerges as pivotal in Mid-zone during early acute injury.

(A) Heatmap displays the area under the curve (AUC) scores of transcription factor (TF) motifs, estimated per dot by Single-Cell Regulatory Network Inference and Clustering (SCENIC), highlighting gene regulatory networks and differentially activated TF motifs in hepatocyte zones (PP, Mid, PC) at 6 h post-APAP. Columns represent TF motifs, rows represent dots, and color intensity indicates AUC scores. (B) Heatmap shows the activity of transcription factors (TFs) across different zonal regions at 6 h post-APAP. TF activity was quantified using AUCCell scoring. The color key represents regulon activity scores. (C) Violin plots show transcriptional activity of Atf4 and Ddit3 in each zonation at 6 h post-APAP. (D) GO pathway analysis reveals the top 15 enriched pathways for genes co-regulated by Atf4 and Ddit3 in mid hepatocyte zones at 6 h post-APAP. (E) Heatmap showing the target genes expression of Atf4 (left) and Ddit3 (right) in hepatocyte zones (zones PP, Mid, and PC) at 6h post-APAP. (F) Immunohistochemistry staining of p-eIF2A in liver sections from mice at 0 h and 3 h post-APAP. p-eIF2A-positive hepatocytes are indicated by red arrows. The number of p-eIF2A-positive cells per field of view (FOV) is quantified. $n=3$ mice/group. (G) Heatmap displays stress response and ER stress-related genes from differentially expressed genes (DEGs) identified in Mid hepatocyte zones at 0 and 6 h post-APAP. Data are represented as means $\pm$ SD; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$; ns, not significant.

cellular stresses that activates the unfolded protein response (UPR) via PERK/IRE1/ATF6. Their overlap occurs at PERK-eIF2 α -ATF4, but ER stress uniquely engages IRE1/XBP1 and ATF6 for ER-specific repair.¹⁵ To evaluate the involvement of ER stress in Mid-zone, we further analyzed the expression levels of genes related to stress response and ER stress based on DEGs identified in Mid-zone at 0, 3, and 6 hours post-APAP treatment (**Figure 4G** and Figure S3G). Most of stress response-related genes are highly upregulated (**Figure 4G** and Figure S3G). However, despite the high expression of *Atf4* and *Ddit3*, other ER stress-related genes, such as *Eif2ak3*, *Atf6*, and *Xbp1*²⁸ either remained unchanged or were downregulated at 3 and 6 hours compared to 0 hour post-APAP (**Figure 4G** and Figure S3G). This analysis highlights the unique role of the Atf4-Ddit3 axis in Mid-zone during early AILI.

To validate Mid-zone responses and transcriptional changes during early AILI, we analyzed an additional scRNA-seq dataset from AILI mice.²⁹ UMAP visualization demonstrated that hepatocytes could be clustered into three distinct zones (PP, PC, and Mid) at 3 and 6 hours post-APAP (Figures S4A and S4B). The zonal gene expression patterns were confirmed, with PC-specific genes (*Glul*, *Cyp2e1*, *Cyp2a5*), PP-specific genes (*Alb*, *Cyp2f2*, *Sds*), and Mid-zone genes (*Hmox1*, *Atf3*, *Ubc*) exhibiting distinct spatial distributions (Figure S4C). GO pathway analysis revealed the top 10 enriched pathways for upregulated genes in Mid-zonal hepatocytes at 3 and 6 hours post-APAP, highlighting key stress-response and cell death processes (Figures S4D and S4E). Consistent with our findings, SCENIC analysis also identified the activation of similar TF motifs across zonal regions, with distinct *Ddit3* and *Atf3* motifs activated in Mid-zone hepatocytes at 3 and 6 hours post-APAP (Figures S4F and S4G). These results reinforce the regulatory changes observed in our ST data, further emphasizing the unique role of mid-zonal hepatocytes in stress adaptation.

The Atf4-Ddit3 axis protects hepatocytes from liver injury

To investigate the role of Atf4-Ddit3 axis in cell fate determination, we first examined Atf4 expression in liver sections by immunohistochemical staining at 0 and 6 hours post-APAP. The results showed gradual increased expression of Atf4 in Mid-zone during early AILI (**Figure 5A**). Subsequently, we employed the adeno-associated virus 8 (AAV8) and thyroxine-binding globulin (TBG) promoter method to overexpress *Atf4* specifically in hepatocytes (**Figure 5B**). Atf4 overexpression significantly reduced AILI, as evidenced by H&E staining and serum ALT levels (**Figure 5C** and **5D**). Furthermore, immunohistochemical staining of Atf4 indicated its localization in hepatocyte nuclei around the injured area (**Figure 5E**). Immunofluorescence and immunohistochemical staining for Ki67 and p-eIF2 α , respectively, showed that Atf4 overexpression notably decreased Ki67 expression and increased p-eIF2 α expression (**Figure 5F** and **5G**). Together, these findings suggest that elevated Atf4 expression enhances ISR, offering stress protection under conditions of reduced proliferation.

The Atf4-Ddit3 axis protects the liver by pausing proliferation via Btg2

Atf4 often works with Ddit3 to mediate stress response, particularly in the context of the ISR.^{23,24,30} Given the low expression of *Atf4* and high expression of *Ddit3* in the Mid-zone, we performed Cut&Run to identify the genomic regions where *Ddit3* binds during the initiation phases of AILI. This approach was used to investigate how the Atf4-Ddit3 axis regulates the proliferation of Mid-zone hepatocytes. Compared with the input control, we observed heightened *Ddit3*-binding signals within the chromatin domains (**Figure 6A**). We further overlapped 6157 genes where *Ddit3* binds, as identified by Cut&Run, with 47 DEGs specifically in the Mid-zone hepatocytes, identifying 39 overlapping genes termed *Ddit3*-binding DEGs (**Figure 6B**). Integrative Genomics Viewer (IGV) plots illustrate *Ddit3*-binding peaks and input controls at specific genomic sites such as *Btg2*, *Atf3*, *Dnajb1*, and *Egr1* regulatory elements (**Figure 6C**).

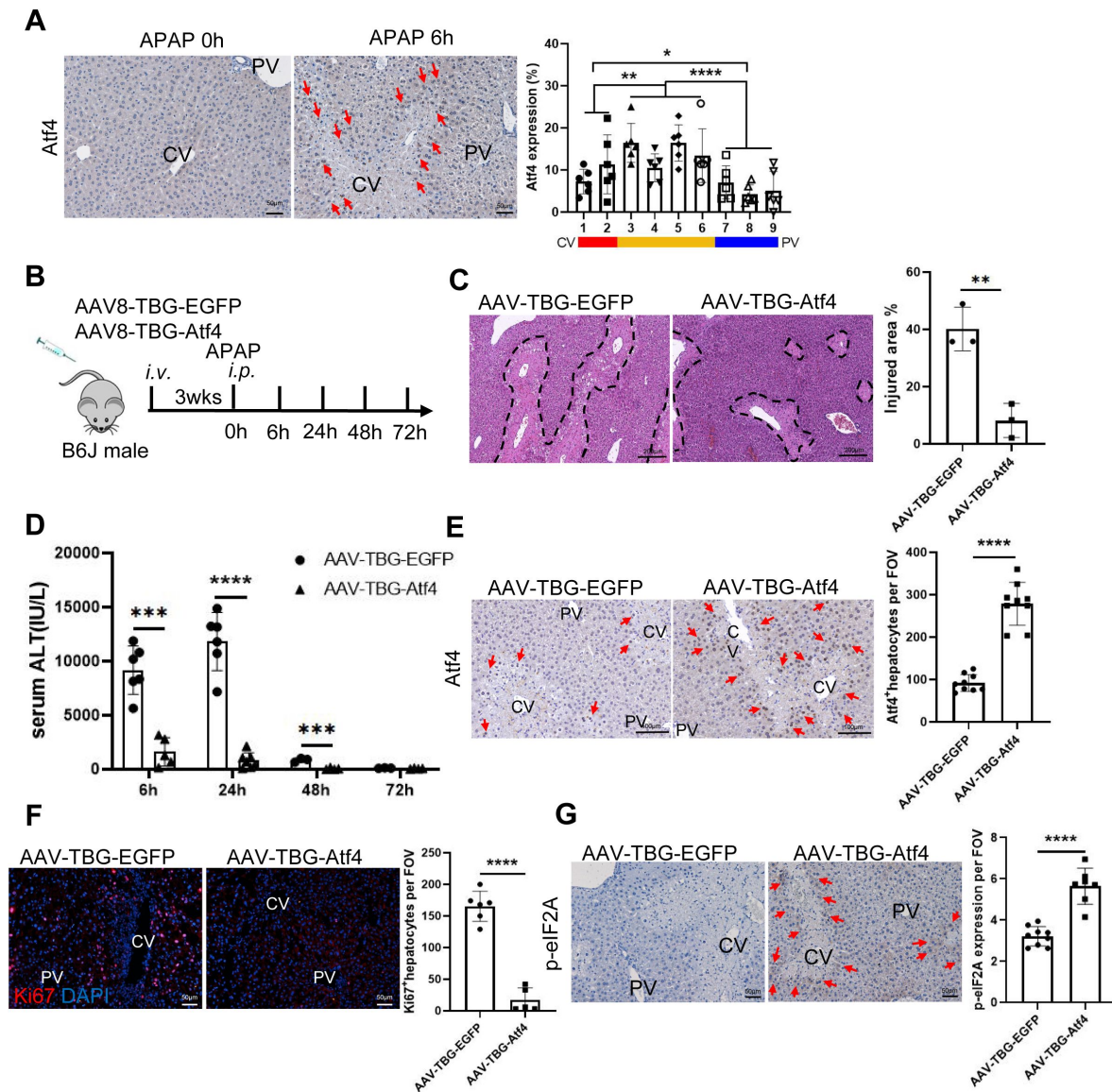


Figure 5

The Atf4-Ddit3 axis protects hepatocytes from liver injury.

(A) Immunohistochemical staining of Atf4 in liver sections at 0 and 6 h post-APAP. Red arrows indicate Atf4-positive hepatocytes. Zonal distribution of Atf4-positive cells in liver sections at 6 h post-APAP is quantified. The statistic is the percentage of Atf4-positive hepatocytes in each layer over the total number of Atf4-positive hepatocytes. $n=3$ mice. (B) Schematic figure illustrating the overexpression of Atf4 via AAV in hepatocytes of wildtype C57BL/6j mice. Overexpression of EGFP is used as a control. (C) H&E staining showing liver morphology from mice overexpressing AAV-TBG-EGFP or AAV-TBG-Atf4 at 6 h post-APAP. The injured area is outlined by black dashed lines. The percentage of injured area is quantified. $n=3$ mice/group. (D) Serum levels of ALT is measured in mice overexpressing AAV-TBG-EGFP or AAV-TBG-Atf4 at 6, 24, 48, and 72 h post-APAP. $n=6$ mice/group. (E) Immunohistochemistry staining of Atf4 in liver sections from mice overexpressing AAV-TBG-EGFP or AAV-TBG-Atf4 at 12 h post-APAP. Atf4-positive hepatocytes are indicated by red arrows. The number of Atf4-positive cells per field of view (FOV) is quantified. $n=9$ mice/group. (F) Immunofluorescence staining of Ki67 (red) was performed to assess proliferating hepatocytes in mice overexpressing AAV-TBG-EGFP or AAV-TBG-Atf4 at 72 h post-APAP. Cell nuclei were stained with DAPI (blue). The number of Ki67-positive cells per FOV is quantified. $n=6$ mice/group. (G) Immunohistochemistry staining of p-eIF2A in liver sections from mice overexpressing AAV-TBG-EGFP or AAV-TBG-Atf4 at 12 h post-APAP. p-eIF2A-positive hepatocytes are indicated by red arrows. The number of Atf4-positive cells per FOV is quantified. $n=6$ mice/group. Data are represented as means $\pm$ SD; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$; ns, not significant.

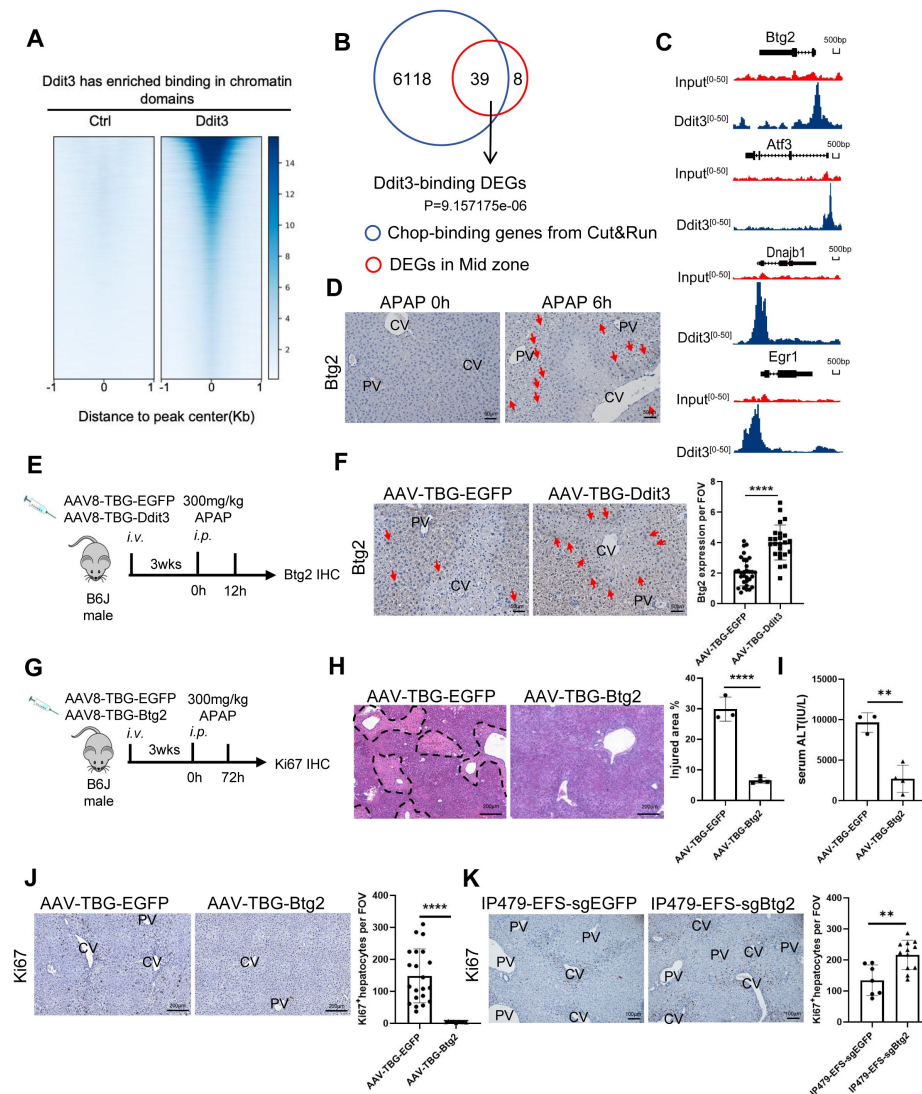


Figure 6.

The Atf4-Ddit3 axis protects the liver by pausing proliferation via Btg2.

(A) Heatmap of Ddit3-binding signals in chromatin domains. (B) Overlap of Ddit3-binding genes by Cut&Run with differentially expressed genes (DEGs) in mid zone was shown. These genes were referred as Ddit3-binding DEGs. (C) Integrative Genomics Viewer (IGV) plot shows the Cut&Run peaks of Ddit3 and input on Btg2-site, Ifrd1-site, Fos-site and Egr1-site. (D) Immunohistochemical staining of Btg2 in liver sections at 0 and 6 h post-APAP. Red arrows indicate Btg2-positive hepatocytes. (E) Schematic figure illustrating overexpression of Ddit3 via AAV in hepatocytes of wildtype C57BL/6J mice. Overexpression of EGFP is used as a control. (F) Immunohistochemical staining of Btg2 in liver sections from mice overexpressing either AAV-TBG-EGFP or AAV-TBG-Ddit3 at 12 h post-APAP. Btg2-positive hepatocytes are indicated by red arrows. Btg2-positive hepatocytes in liver sections is quantified. $n=4$ mice/group. (G) Schematic figure illustrating overexpression of Btg2 via AAV in hepatocytes of wildtype C57BL/6J mice. Overexpression of EGFP is used as a control. (H) H&E staining shows morphology of livers from AAV-TBG-EGFP/Btg2 overexpressing mice at 6 h post-APAP. Injured area is outlined by black dashed lines. The percentage of injury area is quantified. $n=3$ mice/group. (I) Serum levels of ALT is measured at 6h post-APAP in AAV-TBG-EGFP/Btg2 overexpressing mice. $n=3$ mice/group. (J) Immunohistochemical staining of Ki67 in liver sections from mice overexpressing either AAV-TBG-EGFP or AAV-TBG-Btg2 at 72 h post-APAP. Ki67-positive hepatocytes are indicated by red arrows. Ki67-positive cells in liver sections is quantified. $n=4$ mice/group. (K) Immunohistochemical staining of Ki67 in liver sections from mice after knocking down Btg2 via overexpressing either IP479-EFS-sgEGFP or IP479-EFS-sgBtg2 at 72 h post-APAP. Ki67-positive hepatocytes are indicated by red arrows. Ki67-positive cells in liver sections is quantified. $n=4$ mice/group. Data are represented as means $\pm$ SD; $*p < 0.01$; $***p < 0.001$; $****p < 0.0001$; ns, not significant.

Among the Ddit3-binding DEGs, we observed B-cell translocation gene 2 (*Btg2*), which is cell cycle inhibitor.^{31–33} Moreover, *Btg2* is one of the target genes that coregulated by Ddit3 and Atf4. Immunohistochemical staining of *Btg2* in liver sections at 0 and 6 hours post-APAP revealed increased expression of *Btg2* around the injured area in response to APAP (**Figure 6D**). Additionally, we investigated whether Ddit3 could upregulate *Btg2* expression in Mid-zone. We overexpressed Ddit3 in the hepatocytes of wildtype C57BL/6J mice using the AAV8 method, with EGFP overexpression serving as a control (**Figure 6E**). Immunohistochemical staining of *Btg2* in liver sections from mice overexpressing either AAV-TBG-EGFP or AAV-TBG-Ddit3 at 12 hours post-APAP indicated that Ddit3 overexpression significantly upregulated *Btg2* expression post-APAP (**Figure 6F**). This result suggests that *Btg2* is a downstream target of Ddit3.

To explore the role of *Btg2* in liver injury and hepatocyte proliferation, we further overexpressed *Btg2* in hepatocytes of wildtype C57BL/6J mice (**Figure 6G**). H&E staining and serum ALT levels indicated that *Btg2* overexpression significantly reduced AILI (**Figure 6H** and **6I**). Ki67 staining further demonstrated that *Btg2* overexpression significantly decreased hepatocyte proliferation (**Figure 6J**). Moreover, we used the CasRx-sgRNA system³⁴ to knock down *Btg2* by overexpressing either IP479-EFS-sgEGFP or IP479-EFS-sg*Btg2*. In contrast to *Btg2* overexpression, *Btg2* knockdown resulted in increased Ki67 expression in hepatocytes, suggesting enhanced hepatocyte proliferation (**Figure 6K**). Together, these findings illustrate Atf4-Ddit3 axis inhibits hepatocytes proliferation in Mid-zone via *Btg2* during early AILI.

Discussion

Our study reveals that during the early phase of AILI, mid-zone hepatocytes undergo a transient proliferation arrest, orchestrated by a distinctive transcriptional program centered on the Atf4-Chop axis. Spatial transcriptomics and functional assays demonstrate that this adaptive response promotes survival by upregulating the cell cycle inhibitor *Btg2*, highlighting a trade-off between cytoprotection and proliferative capacity. Notably, zonal metabolic heterogeneity in APAP processing drives preferential activation of these stress pathways in mid-zone hepatocytes. These findings uncover a critical mechanism by which hepatocytes prioritize stress adaptation before initiating regeneration, offering new insights into the spatiotemporal regulation of liver repair (**Figure 7**).

Our study reveals that mid-zone hepatocytes employ a protective proliferation arrest during early AILI, mediated by the Atf4-Chop axis of the integrated stress response (ISR). While the ISR shares upstream initiators with ER stress (e.g., eIF2 α phosphorylation), its outcomes are context-dependent. ER stress specifically activates the unfolded protein response (UPR) through sensors inositol-requiring enzyme 1 α (IRE1 α), activating transcription factor 6 (ATF6), and protein kinase RNA-like ER kinase (PERK),³⁵ with genetic studies demonstrating their critical roles in AILI – knockout of IRE1 α , X-box binding protein 1 (XBP1), or Chop consistently attenuates liver damage, confirming the UPR's contribution to hepatotoxicity.^{19,36–38} Unlike classical ER stress, we find that mid-zone hepatocytes leverage the Atf4-Chop axis to transiently halt proliferation via *Btg2*, favoring survival. This aligns with emerging evidence that Chop can paradoxically support adaptation in mitochondrial stress,^{30,39} suggesting that its role in fine-tuning of ISR in mammals.

The antiproliferative activity of *Btg2* has been studied through an integrated network of transcriptional and post-transcriptional cell cycle regulatory mechanisms. A study reveals that *Btg2* functions as a molecular bridge between poly(A)-binding protein cytoplasmic 1 (PABPC1) and the CCR4-NOT deadenylase complex subunit CNOT7 (CAF1), facilitating accelerated degradation of cell cycle-related transcripts through enhanced poly(A) tail removal.³¹ At the transcriptional level, *Btg2* exerts specific control over G2/M progression by disrupting the positive feedback loop between cyclin B1 and its transcriptional activator FOXM1 (forkhead box protein M1), thereby

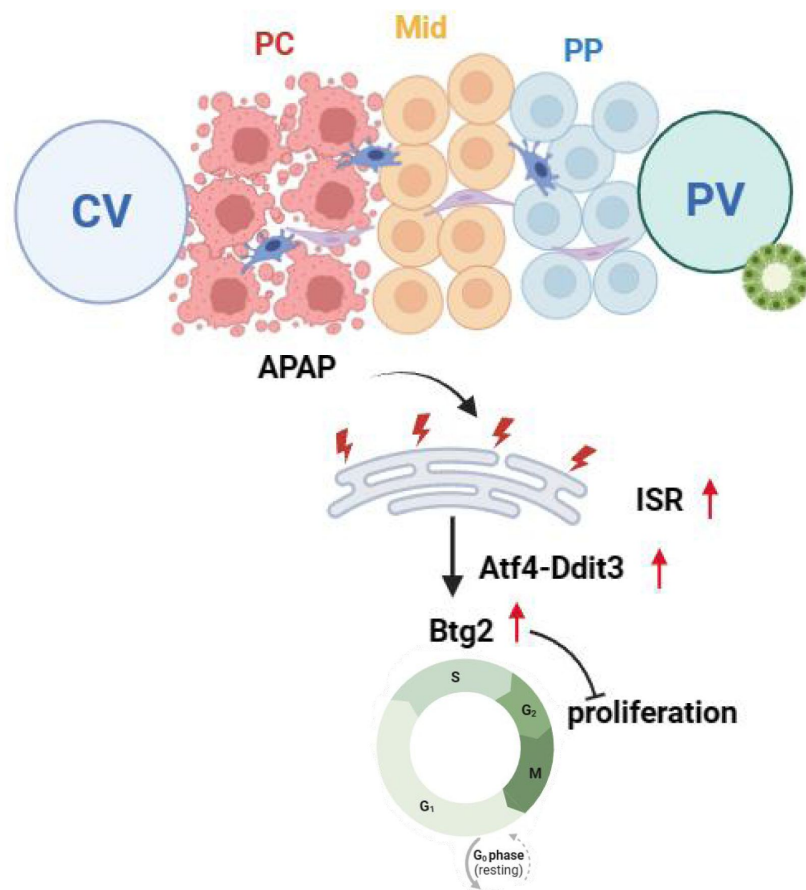


Figure 7.

The Atf4-Ddit3 axis mediates integrated stress protection in mid-zone hepatocytes at the expense of proliferative capacity during early AILI.

Schematic model depicting the adaptive response of mid-zone hepatocytes to acute liver injury, wherein activation of the Atf4-Ddit3 pathway upregulates Btg2 to induce transient cell cycle arrest, balancing survival and regeneration.

suppressing cyclin B1 expression and inducing cell cycle arrest.³² Furthermore, Btg2 demonstrates broader cell cycle inhibitory capacity through its interactions with cyclin-dependent kinases (CDKs) during G1/S transition and its ability to modulate the tumor suppressor p53 (TP53).³³ In the context of AILI, the stress-responsive Atf4-Chop pathway induces Btg2 expression in mid-zonal hepatocytes, where it may coordinate a comprehensive proliferation arrest through these complementary mechanisms. This multilayered regulatory strategy enables hepatocytes to temporarily suspend cell division and redirect resources toward cellular repair and stress adaptation during the acute phase of toxic injury.

A key limitation of this study is the scarcity of early-phase AILI human samples, which currently precludes definitive validation of the clinical relevance of these zonal adaptations. To address this gap, we performed a preliminary analysis of published single-cell RNA sequencing data from APAP-overdose patients.²⁹ Intriguingly, in one of two analyzed patients, mid-zone hepatocytes exhibited transcriptional signatures remarkably consistent with our murine findings, including: (1) upregulation of Atf4-Chop pathways, and (2) downregulation of cell proliferation genes. While the small sample size (n=2) prevents definitive conclusions, these preliminary observations suggest potential conservation of stress-responsive mechanisms between murine and human mid-zone hepatocytes during AILI. Further studies with larger clinical cohorts will be essential to verify these findings and establish their translational significance.

In summary, our study uncovers a spatially coordinated stress adaptation mechanism in mid-zone hepatocytes during early AILI, where the Atf4-Chop axis transiently suppresses proliferation via Btg2 upregulation. By elucidating how hepatocytes prioritize survival before initiating repair, our work provides a framework for understanding the dynamic interplay between stress adaptation and regeneration in tissues.

Methods and materials

Detailed methods can be found in additional files.

Data and code availability

Cut&run data are accessible (GSE272565). ST data are accessible (GSE272564). This paper does not report original code.

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Additional information

Author Contributions

Y.Y.Z. performed the experiments, analyzed the data, and wrote the methods and figure legends. C.X.D. initially analyzed the sequencing data, while B.C. continued the sequencing data analysis, performed the experiments, wrote the methods and figure legends. J.H. prepared the ST samples. Y.N.L. performed the experiments. C.P. supervised the ST data analysis by C.X.D., participated in the initial study design, and revised the manuscript. Z.S. initiated, organized, and designed the study, and wrote the manuscript.

Additional files

Supplementary figures [↗](#)

Supplementary methods [↗](#)

References

- 1 Bernal W., Wendon J (2013) **Acute liver failure** *N Engl J Med* **369**:2525–2534 <https://doi.org/10.1056/NEJMra1208937> | Google Scholar
- 2 Larson A. M., et al. (2005) **Acetaminophen-induced acute liver failure: results of a United States multicenter, prospective study** *Hepatology* **42**:1364–1372 <https://doi.org/10.1002/hep.20948> | Google Scholar
- 3 Ostapowicz G., et al. (2002) **Results of a prospective study of acute liver failure at 17 tertiary care centers in the United States** *Ann Intern Med* **137**:947–954 <https://doi.org/10.7326/0003-4819-137-12-200212170-00007> | Google Scholar
- 4 Lee W. M (2007) **Acetaminophen toxicity: Changing perceptions on a social/medical issue** *Hepatology* **46**:966–970 <https://doi.org/10.1002/hep.21926> | Google Scholar
- 5 Dahlin D. C., Miwa G. T., Lu A. Y., Nelson S. D (1984) **N-acetyl-p-benzoquinone imine: a cytochrome P-450-mediated oxidation product of acetaminophen** *Proc Natl Acad Sci U S A* **81**:1327–1331 <https://doi.org/10.1073/pnas.81.5.1327> | Google Scholar
- 6 Hinson J. A., Reid A. B., McCullough S. S., James L. P (2004) **Acetaminophen-induced hepatotoxicity: role of metabolic activation, reactive oxygen/nitrogen species, and mitochondrial permeability transition** *Drug Metab Rev* **36**:805–822 <https://doi.org/10.1081/DMR-200033494> | Google Scholar
- 7 Jack A., Hinson D. W. R., Laura P. (2010) **James. Mechanism of Acetaminophen-Induced Liver Necrosis** *Handb Exp Pharmacol* :369–405 https://doi.org/10.1007/978-3-642-00663-0_12 | Google Scholar
- 8 Tujios S., Fontana R. J (2011) **Mechanisms of drug-induced liver injury: from bedside to bench** *Nat Rev Gastroenterol Hepatol* **8**:202–211 <https://doi.org/10.1038/nrgastro.2011.22> | Google Scholar
- 9 Ben-Moshe S., et al. (2022) **The spatiotemporal program of zonal liver regeneration following acute injury** *Cell Stem Cell* **29**:973–989 <https://doi.org/10.1016/j.stem.2022.04.008> | Google Scholar
- 10 Wang S. Y., et al. (2024) **Region-specific cellular and molecular basis of liver regeneration after acute pericentral injury** *Cell Stem Cell* **31** <https://doi.org/10.1016/j.stem.2024.01.013> | Google Scholar
- 11 Wu B. H., et al. (2024) **A spatiotemporal atlas of cholestatic injury and repair in mice** *Nat Genet* **56** <https://doi.org/10.1038/s41588-024-01687-w> | Google Scholar
- 12 Wei Y., et al. (2021) **Liver homeostasis is maintained by midlobular zone 2 hepatocytes** *Science* **371** <https://doi.org/10.1126/science.abb1625> | Google Scholar
- 13 He L. J., et al. (2021) **Proliferation tracing reveals regional hepatocyte generation in liver homeostasis and repair** *Science* **371**:905 <https://doi.org/10.1126/science.abc4346> | Google Scholar

- 14 Chembazhi U. V., Bangru S., Hernaez M., Kalsotra A (2021) **Cellular plasticity balances the metabolic and proliferation dynamics of a regenerating liver** *Genome Research* **31**:576–591 <https://doi.org/10.1101/gr.267013.120> | Google Scholar
- 15 Costa-Mattioli M., Walter P (2020) **The integrated stress response: From mechanism to disease** *Science* **368**:384 <https://doi.org/10.1126/science.aat5314> | Google Scholar
- 16 Vasudevan D., et al. (2020) **Translational induction of ATF4 during integrated stress response requires noncanonical initiation factors eIF2D and DENR** *Nature Communications* **11** <https://doi.org/10.1038/s41467-020-18453-1> | Google Scholar
- 17 Marciniak S. J., et al. (2004) **CHOP induces death by promoting protein synthesis and oxidation in the stressed endoplasmic reticulum** *Gene Dev* **18**:3066–3077 <https://doi.org/10.1101/gad.1250704> | Google Scholar
- 18 Han J., et al. (2013) **ER-stress-induced transcriptional regulation increases protein synthesis leading to cell death** *Nat Cell Biol* **15**:481 <https://doi.org/10.1038/ncb2738> | Google Scholar
- 19 Uzi D., et al. (2013) **CHOP is a critical regulator of acetaminophen-induced hepatotoxicity** *Journal of Hepatology* **59**:495–503 <https://doi.org/10.1016/j.jhep.2013.04.024> | Google Scholar
- 20 Huang X., et al. (2023) **S-acylation of p62 promotes p62 droplet recruitment into autophagosomes in mammalian autophagy** *Mol Cell* **83**:3485 <https://doi.org/10.1016/j.molcel.2023.09.004> | Google Scholar
- 21 Richardson R. T., et al. (2006) **Nuclear autoantigenic sperm protein (NASP), a linker histone chaperone that is required for cell proliferation** *Journal of Biological Chemistry* **281**:21526–21534 <https://doi.org/10.1074/jbc.M603816200> | Google Scholar
- 22 Peters J. M., McKay R. M., McKay J. P., Graff J. M (2000) **Casein kinase I transduces Wnt signals** *Dev Biol* **222**:228–228 <https://doi.org/10.1006/devbi.2000.2222> | Google Scholar
- 23 Woo C. W., et al. (2009) **Adaptive suppression of the ATF4-CHOP branch of the unfolded protein response by toll-like receptor signalling** *Nat Cell Biol* **11**:1473–U1203 <https://doi.org/10.1038/ncb1996> | Google Scholar
- 24 Ohoka N., Yoshii S., Hattori T., Onozaki K., Hayashi H (2005) **a novel ER stress-inducible gene, is induced via ATF4-CHOP pathway and is involved in cell death** *Embo J* **24**:1243–1255 <https://doi.org/10.1038/sj.emboj.7600596> | Google Scholar
- 25 Bunch H., et al. (2023) **ERK2-topoisomerase II regulatory axis is important for gene activation in immediate early genes** *Nature Communications* **14** <https://doi.org/10.1038/s41467-023-44089-y> | Google Scholar
- 26 von Scheidt M., et al. (2021) **Transcription Factor MAFF (MAF Basic Leucine Zipper Transcription Factor F) Regulates an Atherosclerosis Relevant Network Connecting Inflammation and Cholesterol Metabolism** *Circulation* **143**:1809–1823 <https://doi.org/10.1161/Circulationaha.120.050186> | Google Scholar

- 27 Lin Y. N., et al. (2024) **Nuclear factor interleukin 3 and metabolic dysfunction-associated fatty liver disease development** *Communications Biology* **7** <https://doi.org/10.1038/s42003-024-06565-z> | Google Scholar
- 28 Zhang J., et al. (2022) **Endoplasmic reticulum stress-mediated cell death in liver injury** *Cell Death & Disease* **13** <https://doi.org/10.1038/s41419-022-05444-x> | Google Scholar
- 29 Matchett K. P., et al. (2024) **Multimodal decoding of human liver regeneration** *Nature* <https://doi.org/10.1038/s41586-024-07376-2> | Google Scholar
- 30 Kaspar S., et al. (2021) **Adaptation to mitochondrial stress requires CHOP-directed tuning of ISR** *Science Advances* **7** <https://doi.org/10.1126/sciadv.abf0971> | Google Scholar
- 31 Stupfler B., Birck C., Séraphin B., Mauxion F (2016) **BTG2 bridges PABPC1 RNA-binding domains and CAF1 deadenylase to control cell proliferation** *Nature Communications* **7** <https://doi.org/10.1038/ncomms10811> | Google Scholar
- 32 Park T. J., et al. (2008) **TIS21 negatively regulates hepatocarcinogenesis by disruption of cyclin B1-Forkhead box M1 regulation loop** *Hepatology* **47**:1533–1543 <https://doi.org/10.1002/hep.22212> | Google Scholar
- 33 Yuniati L., Scheijen B., van der Meer L. T., van Leeuwen F. N (2019) **Tumor suppressors BTG1 and BTG2: Beyond growth control** *Journal of Cellular Physiology* **234**:5379–5389 <https://doi.org/10.1002/jcp.27407> | Google Scholar
- 34 Zhou H. B., et al. (2020) **Glia-to-Neuron Conversion by CRISPR-CasRx Alleviates Symptoms of Neurological Disease in Mice** *Cell* **181**:590 <https://doi.org/10.1016/j.cell.2020.03.024> | Google Scholar
- 35 Wang X. H., et al. (2006) **Mechanism of arylating quinone toxicity involving Michael adduct formation and induction of endoplasmic reticulum stress** *P Natl Acad Sci USA* **103**:3604–3609 <https://doi.org/10.1073/pnas.0510962103> | Google Scholar
- 36 Nagy G., et al. (2007) **Acetaminophen induces ER dependent signaling in mouse liver** *Archives of Biochemistry and Biophysics* **459**:273–279 <https://doi.org/10.1016/j.abb.2006.11.021> | Google Scholar
- 37 Hur K. Y., et al. (2012) **IRE1 α activation protects mice against acetaminophen-induced hepatotoxicity** *Journal of Experimental Medicine* **209**:307–318 <https://doi.org/10.1084/jem.20111298> | Google Scholar
- 38 Ye H., et al. (2022) **Genetic and pharmacological inhibition of XBP1 protects against APAP hepatotoxicity through the activation of autophagy** *Cell Death & Disease* **13** <https://doi.org/10.1038/s41419-022-04580-8> | Google Scholar
- 39 Kress J. K. C., et al. (2023) **The integrated stress response effector ATF4 is an obligatory metabolic activator of NRF2** *Cell Reports* **42** <https://doi.org/10.1016/j.celrep.2023.112724> | Google Scholar

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Reviewer #1 (Public review):

Summary:

The authors present evidence that during acetaminophen (APAP)-induced liver injury, mid-zone hepatocytes activate an integrated stress response (ISR) program via Atf4 and Chop, leading to induction of Btg2. This program suppresses proliferation in the early phase of injury, prioritizing hepatocyte survival before regeneration begins. The study uses spatial transcriptomics, immunohistochemistry, CUT&RUN, and AAV overexpression to support this model.

Strengths:

- (1) Innovative use of spatial transcriptomics to capture zonal differences in hepatocyte stress responses.
- (2) Identification of a mid-zone specific ISR signature and candidate downstream regulator Btg2.
- (3) Functional experiments with Atf4-Chop-Btg2 modulation provide causal evidence linking ISR activation to proliferation inhibition.
- (4) Conceptually significant model that hepatocytes actively balance survival and regeneration dynamically in a zone-specific manner.

Weaknesses:

- (1) Zonation definition under injury has been shown to be sustained broadly, but is not sufficiently validated and quantified, especially considering the resolution of the 10x Visium system and the potential variation of outcomes based on how to define zones.
- (2) The model is built entirely in APAP injury, which specifically targets pericentral hepatocytes. It remains unclear whether the proposed mechanism applies to other liver injuries (e.g., partial hepatectomy, CCl4).
- (3) Baseline proliferation appears higher than expected in homeostasis (Figure 1B), and fold change analysis (not absolute counts) may be needed to assess zonal proliferation suppression (Figure 1D).
- (4) AAV-based overexpression raises potential confounds (altered CYP activity before injury) and shows incomplete penetrance that is not quantified. (Figure 5 - Figure 6).
- (5) The functional link between proliferation suppression and improved survival is inferred, but direct survival /injury readouts are limited.

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Reviewer #2 (Public review):

The manuscript reports protection of midlobular hepatocytes from APAP toxicity by activation of Atf4-CHOP (Ddit3)-mediated cell cycle arrest and stress response. The authors acknowledge that their finding is unexpected because CHOP typically induces cell death. Therefore, they functionally validate several aspects of the proposed Atf4-CHOP mechanism. Along these lines, the mitigation of APAP toxicity by AAV expression of Atf4 or Btg2, the latter identified as CHOP effector, is impressive. Whether Atf4 indeed acts through CHOP and whether midlobular hepatocytes are protected because of cell cycle arrest is less clear. These and other criticisms are described in the following.

Major points:

- (1) Starting with the basics, one wonders why midlobular hepatocytes manage to mount a defensive response to APAP but pericentral hepatocytes don't. Is this because midlobular hepatocytes express the relevant Cyps (2e1, but also 1a2 and 3a11) at lower levels, which mitigates toxicity and buys them time? This would be supported by F2A but not by F3B, at least not for the most important Cyp2e1. A moderate difference is shown for Cyp1a2 expression in F3D, but is that enough to explain the different fates? Or are additional post-transcriptional effects on these Cyps at work?
- (2) The evidence presented in support of cell cycle arrest of midlobular hepatocytes is not fully convincing: there is no overt difference in S and G2/M gene scores in F2F; the marker genes used for S phase and G1 to S progression in F2G are unusual. Along these lines, one wonders if spatial transcriptomics confirmed the Ki67 immunostaining results in F1 also for specific zones, not only overall, as shown in F2E?
- (3) The authors conclude in line 364 that halting of proliferation by Btg2 favors survival, which raises the question of whether Btg2 knockout causes death in midlobular hepatocytes in F6K. Data addressing this question, that is, the localization and extent of tissue necrosis and ALT levels after APAP, are missing. The efficiency of the knockout of Btg2 is also not given.
- (4) Related to the previous question, the BTG2 immunostaining in F6F is not convincing when compared to F6D. One also wonders if it is necessary to apply APAP to find induction of BTG2 by AAV-Ddit3?
- (5) Related to the previous question, the proposed Atf4-Ddit3 axis is challenged by the lack of midlobular induction of Atf4 in the APAP scRNA-seq data published by another group, presented in S4F and G. Further analysis of AAV-Atf4 samples generated for F5 could address whether it is really Atf4 that acts on Ddit3 in APAP toxicity.
- (6) Related to the previous question, the ATF4 immunostaining in F5A doesn't look convincing, with many brown pigments appearing to be outside of the nucleus.
- (7) It is not ruled out that AAV expression of Atf4 or Btg2 reduces hepatocyte sensitivity to APAP by affecting the expression of the Cyps needed for activation. In other words, does AAV-Atf4 or AAV-Btg2 change the expression of any of the Cyps relevant to APAP in the 3 weeks before APAP application (F5B)?
- (8) It is laudable that the authors tried to extend their findings to humans by using snRNA-seq data from a published study (line 391), but it is unclear why they didn't analyze all 10 patients in that study but instead focused on 2 and stated that this small sample number prevented drawing definitive conclusions and could therefore only be mentioned in the discussion.

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Reviewer #3 (Public review):**Summary:**

This paper by Zhu et al explores zonal gene expression changes and stress responses in the liver after APAP injury. 3-6 hours after APAP, zone 2 hepatocytes demonstrate important gene expression changes. There is an increase in stress response/cell survival genes such as Hmox1, Hspa8, Atf3, and protein degradation/autophagy genes such as Ubb, Ubc, and Sqstm1. This is hypothesized to be a "stress adaption" which happens during the initial phases of acute liver injury. Furthermore, there is a spatial redistribution of Cyp450 expression that then establishes the Mid-zone as the primary site of APAP metabolism during early AILI. This particular finding was identified previously by other groups in several single-cell papers. Ddit3 (Chop) expression also increases in zone 2. The authors focused mostly on the Atf4-Ddit3 axis in stress adaptation. Importantly, they probe the functionality of this axis by overexpressing either ATF4 or DDIT3 using AAV tools, and they show that these manipulations block APAP-induced injury and necrosis. This is somewhat convincing evidence that these stress response proteins are probably important during injury and regeneration.

Strengths:

Overall, I think this is a useful study, showing that the Mid-lobular zone 2 hepatocytes turn on a stress-responsive gene program that suppresses proliferation, and that this is functionally important for efficient, long-term regeneration and homeostasis. This adds to the body of literature showing the importance of zone 2 cells in hepatic regeneration, and also provides an additional mechanism that tells us how they are better at surviving chemical injuries.

Weaknesses:

The main concern is that the overexpression of ATF4 and DDIT3 is causing reduced cell death and damage by APAP. This makes it harder to understand if these genes are truly increasing survival or if they are just reducing the injury caused by APAP. It may be better to perform overexpression immediately after, or at the same time as APAP delivery. Alternatively, loss-of-function experiments using AAV-shRNAs against these targets could be useful.

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